

Written exam 19 December, 2018

Course title: Network Optimization

Course no: 42115

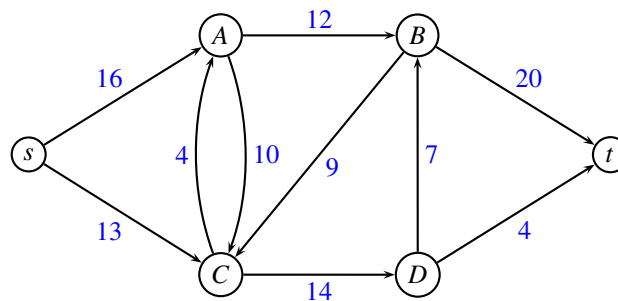
Aids allowed: All

Exam duration: 4 hours

Weighting: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 40 points can be obtained.

Language: You can write in Danish or English.

Maximum Flow problem



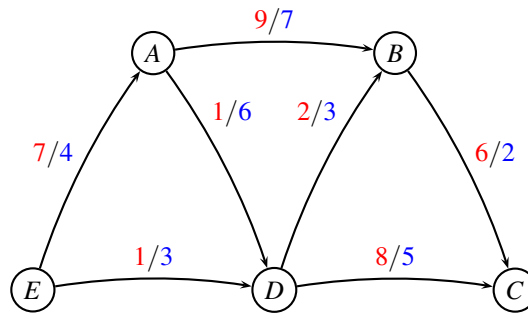
Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration report which augmenting path was used and the capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 2: (medium) Draw the residual graph when the algorithm terminates. Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Empty container repositioning

Consider the following network of a small shipping company, MEDITLINE. Nodes correspond to ports, and edges correspond to connections. The cost of shipping one container is indicated with **red**, while the capacity of the connection is indicated with **blue**.

MEDITLINE: Route net of a Mediterranean liner shipping company



node	A	B	C	D	E
demand	-2	4	3	0	-5

Due to the imbalance in trade, empty containers tend to accumulate at various ports, while other ports have a deficit. For MEDITLINE, the demands for empty containers are stated in the table next to the network. Positive values denote a demand, while negative values denote a surplus of containers. It is assumed that all containers are identical.

Problem 3: (small) Describe how you can formulate the repositioning problem as a network problem, and state which algorithm you will use to solve it. ■

Problem 4: (large) Solve the MEDITLINE repositioning problem. Give sufficient details to make it possible to follow your calculations. Report the optimal solution and the cost of the solution. ■

Critical Path problem

Ship's biscuit was a hard piece of bread that sailors ate at nearly every meal. The biscuit was baked on land, stored on board the ship, and then sent out to sea with the sailors. Sailors soaked the rock-hard biscuit in their stew to soften it before taking a bite.



In order to bake ship's biscuit one has to go through the below list of activities. Each activity has a duration and some predecessor tasks that need to be finished before the activity can start.

Activity	Description	Predecessors	Duration
A	Start oven at 250 degrees	D, B	8
B	Weigh rye flour and wheat flour		9
D	Dissolve yeast in the water		7
C	Prepare a barrel for storage	E	5
E	Cut the dough into squares	A	9
F	Bake the biscuits	G, E	6
G	Mix ingredients and add salt	D	12

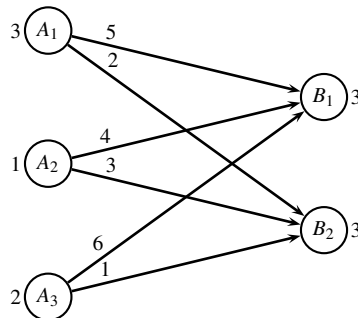
Problem 5: (small) Order the activities in topological order. ■

Problem 6: (large) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan. ■

Problem 7: (small) Mark the critical path. ■

Fixed charge transportation problem

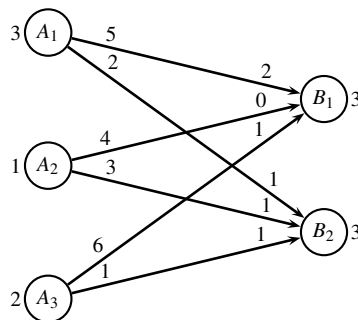
Three factories A_1, A_2, A_3 are transporting items to two customers B_1, B_2 . The *transportation costs* of sending one item are indicated on the graph below. The capacity of the factories is respectively $p_1 = 3, p_2 = 1, p_3 = 2$, while the demand at the customers is $d_1 = 3, d_2 = 3$. There is a *fixed charge* of cost 4 associated with each transportation. The fixed charge is to be paid if one or more items are sent along an edge, however, the fixed charge is independent of the number of items sent.



Due to the fixed charge constraints, we use a Dantzig-Wolfe decomposed formulation, where we write up all patterns of sending the p_i items from node A_i . The cost of a pattern consists of the transportation costs, and the fixed charges for using edges.

Problem 8: (medium) Write up the Dantzig-Wolfe decomposed model. The first constraint should make sure that the d_1 items arrive at customer B_1 , and the second constraint should make sure that the d_2 items arrive at customer B_2 . ■

Since the Dantzig-Wolfe decomposed model may contain many columns for more complex problems, we use delayed column generation to solve the LP-relaxed master problem. We start with the initial flow:

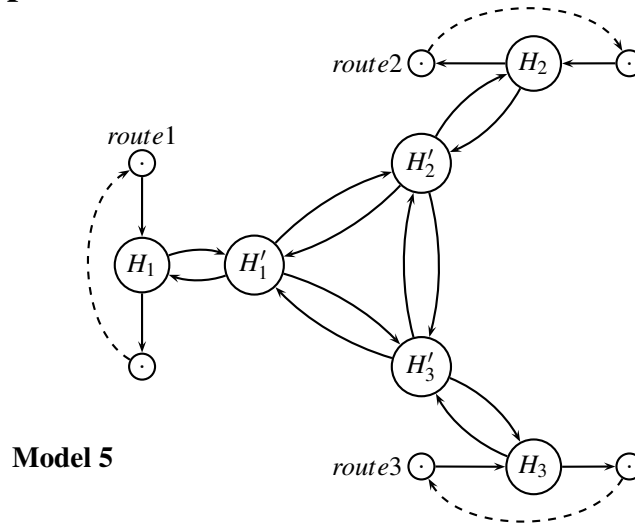


(numbers at start of edge denote costs, numbers at end of edge denote flow). The restricted master problem is now solved. Let u_i be the dual variables corresponding to the constraints saying that d_i items arrive at customer B_i . Let v_j be the dual variables corresponding to the constraints saying that only one pattern may be used for factory A_j .

The dual variables for the restricted master problem are $u_1 = 0, u_2 = 0, v_1 = 20, v_2 = 7, v_3 = 15$.

Problem 9: (medium) Calculate the reduced costs of all patterns (that are not currently in the restricted master problem). Which pattern should be added to the restricted master problem (and why?). ■

Modeling a hub port



Consider Model 5 from Project Assignment 2018. The graph is used to model a hub port, where three routes meet and transship containers between the vessels. We model the hub port H by using an extra node H_1 , H_2 and H_3 for each route, and additional nodes H'_1 , H'_2 and H'_3 . Edges are added between all pairs of nodes H'_1 , H'_2 , and H'_3 .

Problem 10: (medium) Assume that two vessels (route1 and route2) arrive the same day. Therefore, they have to share the crane capacity for loading and unloading. This means that the number of containers unloaded from vessels at route1 and route2 in total should not exceed $u = 8$. Similarly, the number of containers loaded to the two routes together should not exceed $u = 8$. There is sufficient time to transship containers between route1 and route2 during the same day. Modify the above graph to handle these constraints. Indicate capacities on edges where relevant. The resulting graph should still be a multicommodity flow problem. ■

Flooding

Network flow issues come up in dealing with natural disasters and other crises, since major unexpected events often require the movement and evacuation of large numbers of persons in a short amount of time.

Consider the following scenario: Due to large-scale flooding in a region, paramedics have identified a set of n injured persons distributed across the region who need to be rushed to hospitals. There are k hospitals in the region, and each of the n persons needs to be brought to a hospital that is within a half-hour's driving time of their current location (so different persons will have different options for hospitals, depending on where they are right now).

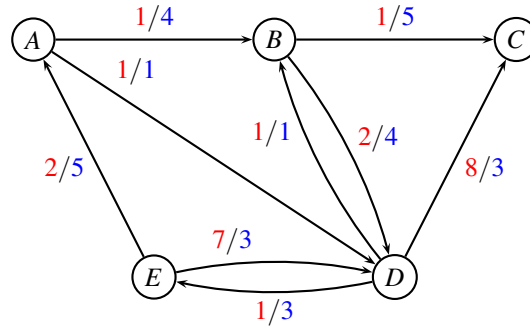
At the same time, one doesn't want to overload any one of the hospitals by sending too many patients its way. The paramedics are in touch by cell phone, and they want to collectively work out whether they can choose a hospital for each of the injured persons in such a way that the load on the hospitals is balanced: Each hospital receives at most $\lceil n/k \rceil$ persons.

Problem 11: (large) Given information about the persons's locations, we would like to know whether the assignment is possible. Formulate the problem as a network optimization problem that can be solved in polynomial time. ■

Unsplittable multicommodity flow problem

We are considering the *unsplittable multicommodity flow problem* from Project Assignment 2018. We have a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost $c_{ij} \geq 0$ and a capacity $u_{ij} \geq 0$. Moreover we are given a number of commodities $K = \{1, \dots, m\}$ each given by a triple (s_k, t_k, d_k) , where s_k denotes the source of commodity k , t_k denotes the target of commodity k , and d_k denotes how many units to be sent from s_k to t_k without splitting the cargo.

ASIANNET (Route net of a liner shipping company)



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij} . Cities are $A = \text{Manilla}$, $B = \text{Shanghai}$, $C = \text{HongKong}$, $D = \text{Xiamen}$, $E = \text{Singapore}$.

Like in the project assignment, we have three commodities with the following demand:

k	s_k	t_k	d_k
1	E	D	2
2	D	C	2
3	E	C	1

Let R_k denote the set of possible routes from s_k to t_k allowing d_k units of capacity. Let moreover $a_{r,ij}^k \in \{0, 1\}$ denote if route r for commodity k is using edge (i, j) , and let $\hat{c}_r^k$ denote the cost for sending d_k containers along route $r \in R_k$. Let the binary variable x_r^k for each route $r \in R_k$ indicate whether or not the route is used for commodity k .

We can now write up the *path formulation* of the *unsplittable multicommodity flow problem*:

$$\min \sum_{k \in K} \sum_{r \in R_k} \hat{c}_r^k x_r^k \quad (1)$$

$$\text{s.t.} \sum_{k \in K} \sum_{r \in R_k} d_k a_{r,ij}^k x_r^k \leq u_{ij} \quad (i, j) \in E \quad (2)$$

$$\sum_{r \in R_k} x_r^k \geq 1 \quad k \in K \quad (3)$$

$$x_r^k \in \{0, 1\} \quad k \in K, r \in R_k \quad (4)$$

Constraint (2) ensures that the capacity of each edge is respected, while (3) ensures that at least one route from s_k to t_k is selected. Finally, (4) ensures that the demand d_k is not split.

We solve the LP-relaxed path formulation of ASIAnET, through column generation, starting with the following routes:

commodity k	route	cost
1	EABD	10
2	DC	16
3	EADBC	5
3	EDC	15

This leads to the following *restricted master problem*:

$$\begin{array}{llllllll} \min & 10x_1 & + & 16x_2 & + & 5x_3 & + & 15x_4 \\ \text{s.t.} & 2x_1 & & & & & & \leq 4 & (y_{AB}) \\ & & & x_3 & & & & \leq 1 & (y_{AD}) \\ & & & x_3 & & & & \leq 5 & (y_{BC}) \\ & 2x_1 & & & & & & \leq 4 & (y_{BD}) \\ & & & x_3 & & & & \leq 1 & (y_{DB}) \\ & & 2x_2 & + & x_4 & & & \leq 3 & (y_{DC}) \\ & & & & & & & \leq 3 & (y_{DE}) \\ & & & x_3 & & & & \leq 5 & (y_{EA}) \\ & 2x_1 & & + & x_4 & & & \leq 3 & (y_{ED}) \\ & x_1 & & & & & & \geq 1 & (\bar{y}_1) \\ & & x_2 & & & & & \geq 1 & (\bar{y}_2) \\ & & & x_3 & + & x_4 & & \geq 1 & (\bar{y}_3) \\ & 0 \leq x_1, x_2, x_3, x_4 \leq 1 \end{array} \quad (5)$$

The dual solution to the model is: $y_{AD} = -10$, $\bar{y}_1 = 10$, $\bar{y}_2 = 16$, $\bar{y}_3 = 15$, while all other dual variables are zero.

Problem 12: (medium) Formulate the problem of finding a column with most negative reduced cost as a series of shortest path problems. For each commodity draw the graph and describe how you would solve the pricing problem. ■

Problem 13: (large) Find the column with most negative reduced cost, and add it to the above model. Give sufficient details to follow your calculations. ■